
ADR-005

Translation Layer Architecture

C3 Contract Control Center

Architecture Decision Record

Status: Approved

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Decision ID: ADR-005

Purpose

This Architecture Decision Record defines the Translation Layer used by the C3 Contract Control Center to isolate the application's domain model from SharePoint-specific implementation details.

The objective is to ensure that external storage schemas never influence the internal business model, preserving a clean separation between infrastructure and application logic.

Context

SharePoint returns data using its own storage model.

Examples include:

- Internal column names
- Lookup objects
- Person fields
- Choice fields
- Hyperlink fields
- SharePoint metadata

These structures are optimized for storage rather than application development.

Directly exposing SharePoint objects to the user interface or business logic would tightly couple the application to SharePoint, making future schema changes difficult and increasing maintenance costs.

To prevent this, C3 introduces a dedicated Translation Layer.

Decision

All external data entering the application shall be translated into domain DTOs before being consumed by business logic or presentation components.

The Translation Layer shall be the only application layer permitted to understand SharePoint internal field names.

Approved Architecture

Every data request follows the approved translation pipeline:

SharePoint List

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PnPJS

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Raw SharePoint Item

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Mapper

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Domain DTO

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Business Rules

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React UI

Translation Responsibilities

The Translation Layer is responsible for:

- Mapping SharePoint internal names
- Flattening lookup fields
- Normalizing Person fields
- Converting hyperlink objects
- Handling null values
- Applying deterministic default values
- Producing strongly typed domain objects

No additional business logic shall exist within the mapper.

Current Translation Components

The platform currently includes:

- contractMapper.ts
- personMapper.ts
- amendmentMapper.ts

Each mapper converts SharePoint payloads into the corresponding domain DTO.

Future domains shall introduce their own dedicated mapper implementations.

Allowed Pattern

SharePoint

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Mapper

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Contract DTO

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Business Logic

Prohibited Pattern

The following architecture is explicitly prohibited:

SharePoint

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React Component

Likewise, business rules shall never reference SharePoint internal names such as:

- ContractStage1
- Disposition1
- CurrentGameTitle
- CurrentTeam
- PrimaryDepartment

These details belong exclusively within the Translation Layer.

Benefits

Infrastructure Independence

Changes to SharePoint schemas require updates only within mapper implementations.

The remainder of the application remains unchanged.

Stable Domain Model

The application's DTOs remain the authoritative representation of business entities.

They are not constrained by SharePoint naming conventions or storage decisions.

Improved Maintainability

Schema evolution becomes localized.

Examples include:

- Column renames
- Lookup modifications
- Additional metadata
- New optional fields

Only the corresponding mapper requires modification.

Cleaner Business Logic

Business services operate exclusively on normalized domain objects.

No SharePoint-specific parsing or defensive null handling is required outside the Translation Layer.

Trade-offs

The Translation Layer introduces:

- Additional source files
- Initial mapping effort
- Mapper maintenance

These costs are outweighed by the long-term reduction in coupling and maintenance complexity.

Implementation

Current implementation:

src/mappers/

contractMapper.ts

personMapper.ts

amendmentMapper.ts

index.ts

Additional domains shall follow the same architectural pattern.

Architecture Principle

External storage models are implementation details. The application operates exclusively on domain models.

The Translation Layer exists to preserve this separation.

Decision

Approved

The Translation Layer is adopted as a permanent architectural principle of the C3 Contract Control Center. All current and future external integrations shall implement dedicated translation components before interacting with the application domain model.
